

1. Two particles P and Q have masses m and $4m$ respectively. The particles are at rest on a smooth horizontal plane. Particle P is given a horizontal impulse, of magnitude I , in the direction PQ . Particle P then collides directly with Q . Immediately after this collision, P is at rest and Q has speed w . The coefficient of restitution between the particles is e .

(a) Find I in terms of m and w .

(2)

(b) Show that $e = \frac{1}{4}$

(1)

(c) Find, in terms of m and w , the total kinetic energy lost in the collision between P and Q .

(2)

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1. Two particles, P and Q , of masses $3m$ and $2m$ respectively, are moving on a smooth horizontal plane. They are moving in opposite directions along the same straight line when they collide directly.

Immediately before the collision, P is moving with speed $2u$.

The magnitude of the impulse exerted on P by Q in the collision is $\frac{9mu}{2}$

- (a) Find the speed of P immediately after the collision.

(3)

The coefficient of restitution between P and Q is e .

Given that the speed of Q immediately before the collision is u ,

- (b) find the value of e .

(5)

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Whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Answer ALL questions. Write your answers in the spaces provided.

1. A small ball of mass 0.3 kg is released from rest from a point 3.6 m above horizontal ground. The ball falls freely under gravity, hits the ground and rebounds vertically upwards.

In the first impact with the ground, the ball receives an impulse of magnitude 4.2 N s . The ball is modelled as a particle.

- (a) Find the speed of the ball immediately after it first hits the ground.

(5)

- (b) Find the kinetic energy lost by the ball as a result of the impact with the ground.

(3)

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1. A particle A has mass $2m$ and a particle B has mass $3m$. The particles are moving in opposite directions along the same straight line and collide directly.

Immediately **before** the collision, the speed of A is $2u$ and the speed of B is u .
Immediately **after** the collision, the speed of A is $0.5u$ and the speed of B is w .

Given that the direction of motion of each particle is reversed by the collision,

- (a) find w in terms of u (3)
- (b) find the coefficient of restitution between the particles, (3)
- (c) find, in terms of m and u , the magnitude of the impulse received by A in the collision. (3)

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2. Two particles, A and B , of masses $2m$ and $3m$ respectively, are moving on a smooth horizontal plane. The particles are moving in opposite directions towards each other along the same straight line when they collide directly. Immediately before the collision the speed of A is $2u$ and the speed of B is u . In the collision the impulse of A on B has magnitude $5mu$.

(a) Find the coefficient of restitution between A and B .

(9)

(b) Find the total loss in kinetic energy due to the collision.

(4)

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3. Three particles A , B and C are at rest on a smooth horizontal plane. The particles lie along a straight line with B between A and C .

Particle B has mass $4m$ and particle C has mass km , where k is a positive constant. Particle B is projected with speed u along the plane towards C and they collide directly.

The coefficient of restitution between B and C is $\frac{1}{4}$

- (a) Find the range of values of k for which there would be no further collisions.

(8)

The magnitude of the impulse on B in the collision between B and C is $3mu$

- (b) Find the value of k .

(4)

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4. Two particles, P and Q , have masses m and em respectively. The particles are moving on a smooth horizontal plane in the same direction along the same straight line when they collide directly. The coefficient of restitution between P and Q is e , where $0 < e < 1$

Immediately before the collision the speed of P is u and the speed of Q is eu .

- (a) Show that the speed of Q immediately after the collision is u .

(6)

- (b) Show that the direction of motion of P is unchanged by the collision.

(3)

The magnitude of the impulse on Q in the collision is $\frac{2}{9}mu$

- (c) Find the possible values of e .

(4)

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4. A particle P of mass $3m$ is moving in a straight line on a smooth horizontal floor. A particle Q of mass $5m$ is moving in the opposite direction to P along the same straight line.

The particles collide directly.

Immediately before the collision, the speed of P is $2u$ and the speed of Q is u .

The coefficient of restitution between P and Q is e .

- (a) Show that the speed of Q immediately after the collision is $\frac{u}{8}(9e + 1)$
- (b) Find the range of values of e for which the direction of motion of P is not changed as a result of the collision.

- (b) Find the range of values of e for which the direction of motion of P is not changed as a result of the collision.

When P and Q collide they are at a distance d from a smooth fixed vertical wall, which is perpendicular to their direction of motion. After the collision with P , particle Q collides directly with the wall and rebounds so that there is a second collision between P and Q . This second collision takes place at a distance x from the wall.

Given that $e = \frac{1}{18}$ and the coefficient of restitution between Q and the wall is $\frac{1}{3}$

- (c) find x in terms of d . (6)

